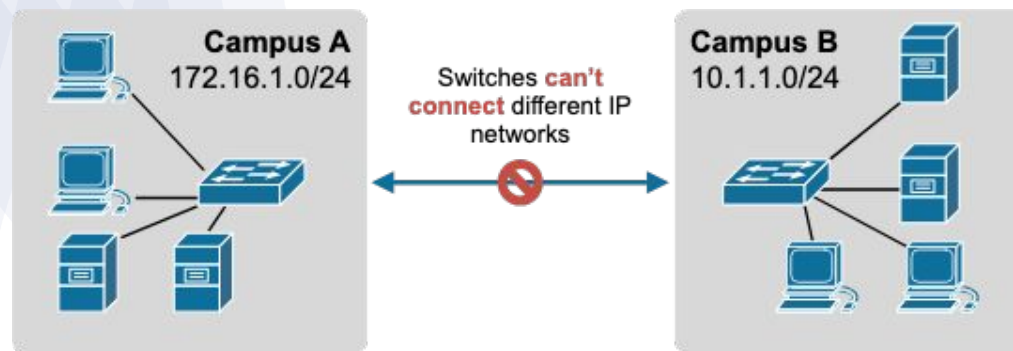




Introduction to Routing

- Early networks were small and local with a few devices. Eventually, **these networks need be connected to each other** to exchange resources. Switches can connect devices within the same network, but they can't connect different IP networks, as shown.
- A switch operates within a single VLAN and broadcast domain, which matches one IP subnet. It uses a flood-and-learn process to flood frames with unknown MAC addresses to all devices within the VLAN.
- A switch does not read the IP header of packets, so it doesn't know the source and destination IP addresses and cannot figure out how to send the packet to different networks or subnets.
- Connecting two separate IP networks requires a device capable of reading and interpreting the IP headers of packets e.g. routers



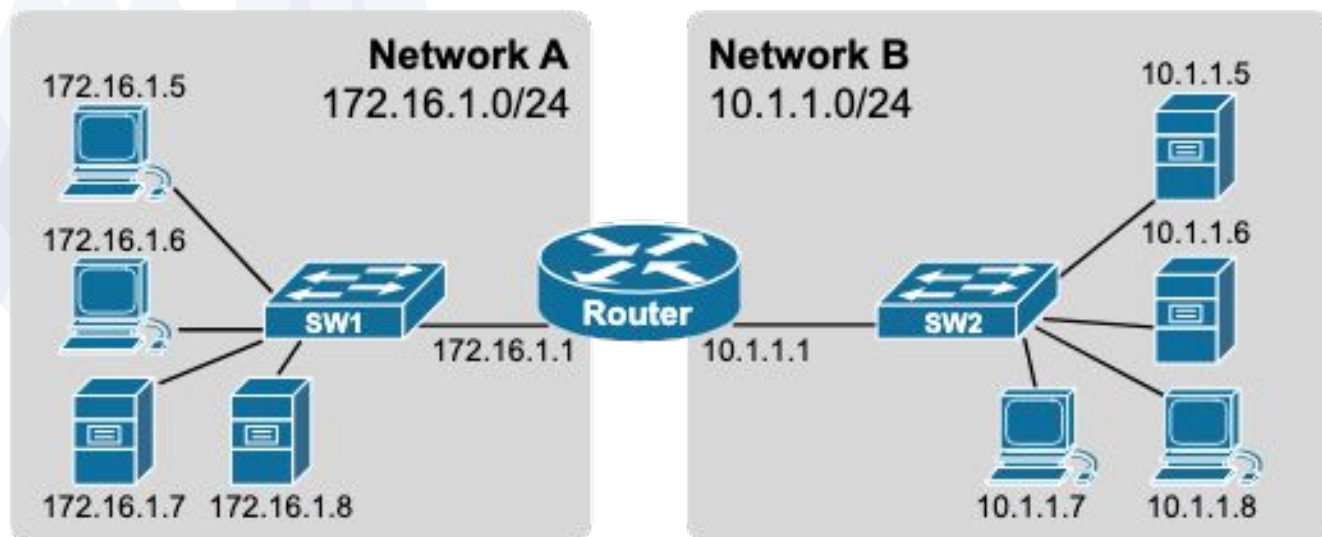
Routers



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- A router connects multiple switches and their networks to create a larger, interconnected IP network and enables communication between IP networks.
- For example, two networks: 172.16.1.0/24 (A) and 10.1.1.0/24 (B). To link them, we use a router with one interface connected to each network, as shown below.
- In the enterprise, routers typically connect different departments, buildings, and remote sites to the data center or the cloud.

NOTE: End devices, like computers and servers, do not connect directly to a router. They connect to a switch, and the switch connects to a router (directly or via another switch).

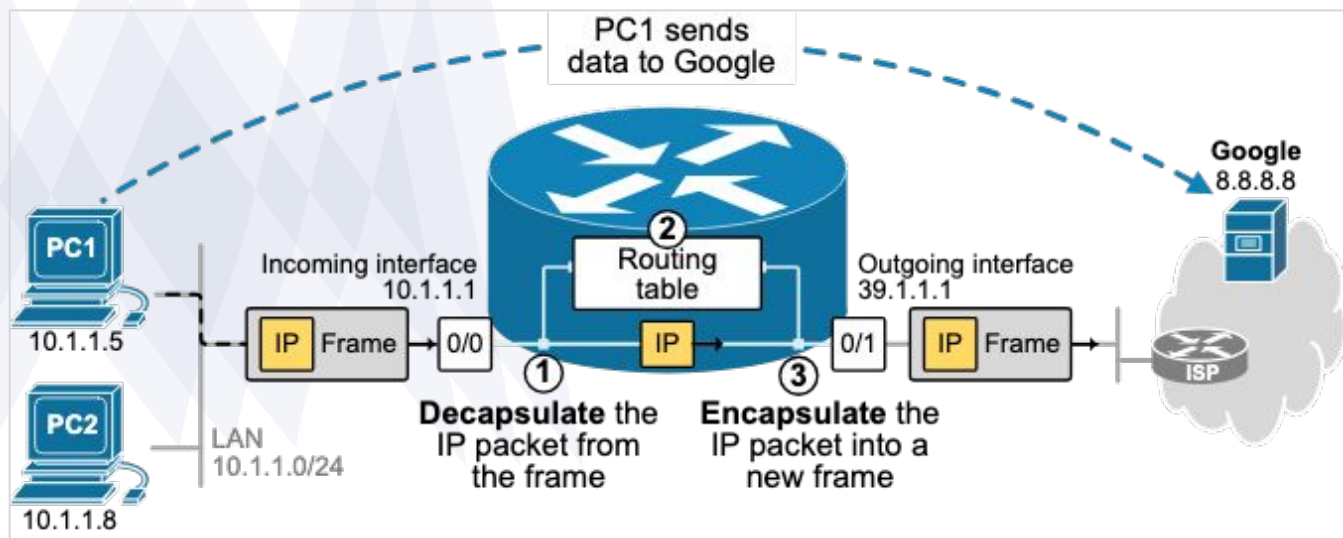


Routers Working



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- A router works by using destination IP addresses to decide where to send data between different networks. Every router has two main components related to how it forwards packets:
 - Interfaces - These are the physical or virtual ports that connect the router to different networks. Each interface belongs to a different subnet and has its own IP address, which is the router's identity in that subnet. It is essential to understand and remember that a router can have only one interface in a specific network because it is the device that separates networks. It cannot have two or more interfaces connected to the same network.
 - Routing table - This is a table stored on the router that tells it where to send packets. Each entry shows a destination network, the next hop (or exit interface), and a metric or cost. The table can be built manually (static routes) or automatically (dynamic routing protocols).

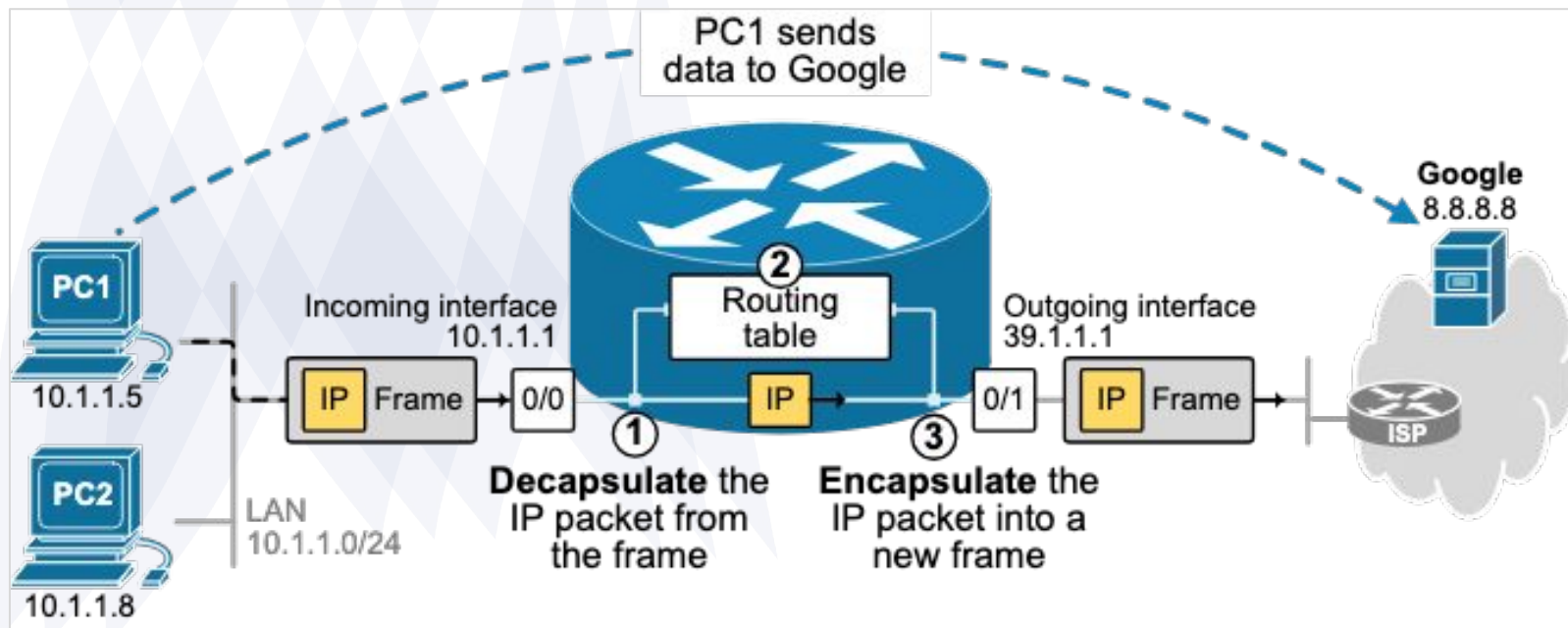


Routers Working



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- Consider what happens when PC1 sends data to Google, as shown.
 - The router receives a frame on the interface connected to the local LAN. It first removes the frame and pulls the IP packet. It then reads the IP header of the packet.
 - Once it knows the destination IP address, it compares it against the entries in its routing table. The routing table tells where to send the packet.
 - Once the router knows which interface it must use to forward the packet to the next hop, it encapsulates it in a new frame destined to the destination MAC address of the next hop device.



What is Routing?



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- **Routing** is the process of forwarding packets from source node to the destination node
- In order to route the best route or path to send the packets must be determined
- This path or route information is stored in a **routing table**
- The best route or path is **computed** by a **routing algorithm**

IP Routing Starts At The Host



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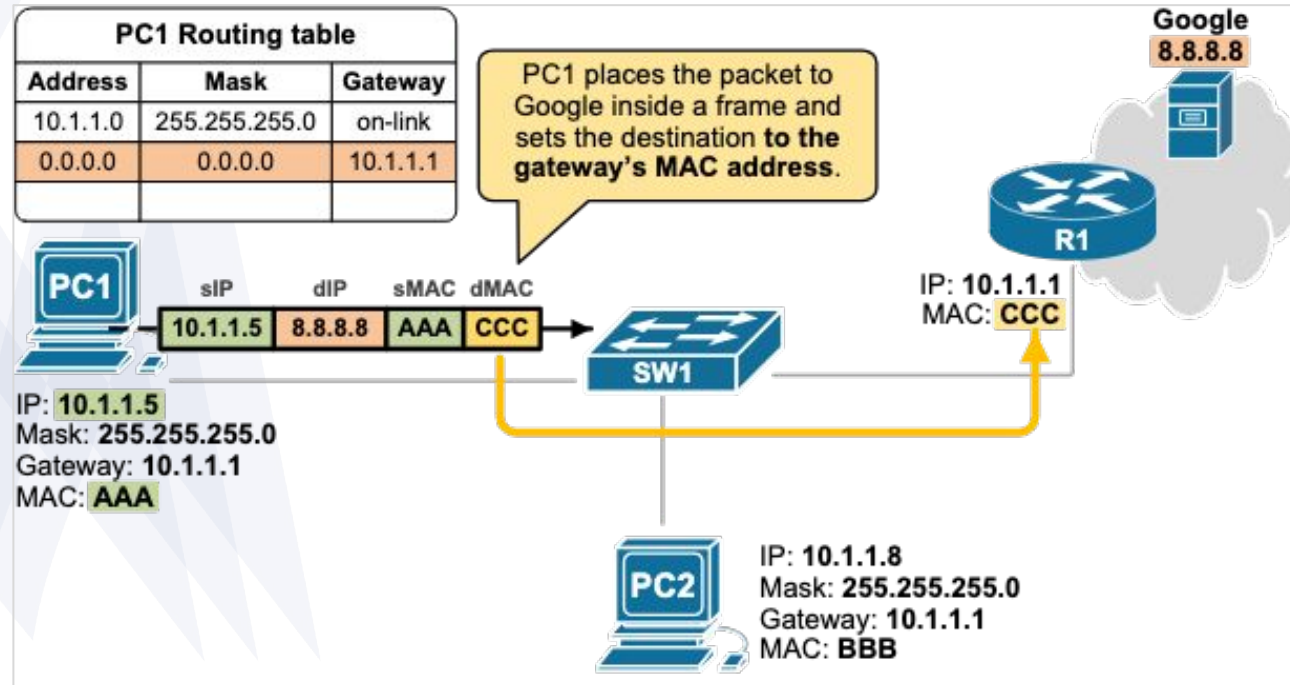
- The very first routing decision happens before the IP packet even gets into the network. The host that originates the data creates an IP packet with a source and destination IP address. While doing it, it uses its own configured IP address, subnet mask, and the destination IP to figure out one of the following two scenarios:
 - If the destination IP is on the same network, it sends the packet directly to that device using ARP to find its MAC address. No router needed.
 - If the destination is on a different network, the host knows it must send the IP packet to its configured default gateway — usually a router connected to the local network. It wraps the packet in a frame addressed to the router's MAC address and places it on the LAN.
- The router only comes into play after the host has already decided that the packet needs to leave its local network.

IP Routing Starts At The Host



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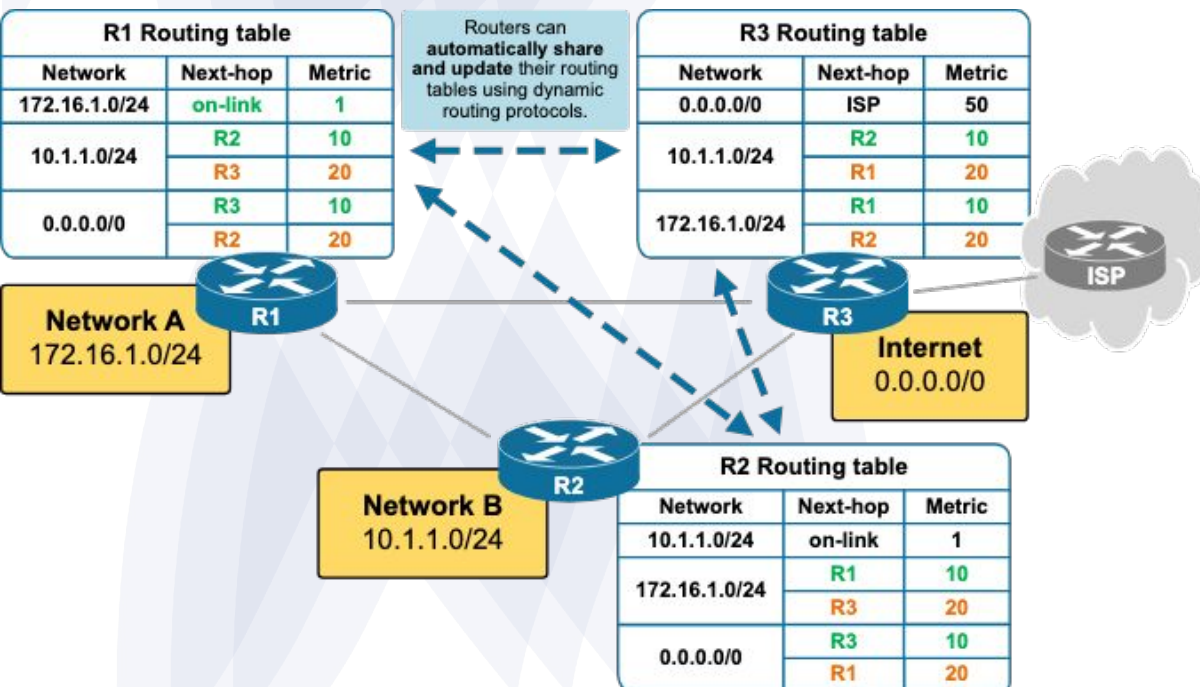
- Using the same example again, but with a focus on PC1.
 - PC1 wants to send data to Google at the IP address 8.8.8.8.
 - It compares the IP 8.8.8.8 with its configured IP and Mask 10.1.1.0 255.255.255.0.
 - Using basic subnetting calculation, it determines that Google's IP is outside its local LAN, so it must send the packet to its configured local gateway router 10.1.1.1.
 - Hence, when PC1 encapsulates the packet into an Ethernet frame, it sets the destination MAC address to R1's MAC address CCC (in yellow).



- The IP packet destined for Google reaches router R1, which performs IP routing and sends it to the next-hop along the way.
- So, even though “routing” is what routers do, hosts have their own routing tables. The host’s routing decision determines whether the packet goes.

IP Routing

- Routers forward traffic according to their IP routing tables. IP routing is the process of moving packets from one network to another using IP addresses as the “destination labels.”
- A router looks at the destination IP in each packet, checks its routing table, and decides the best path to send that packet toward its destination.

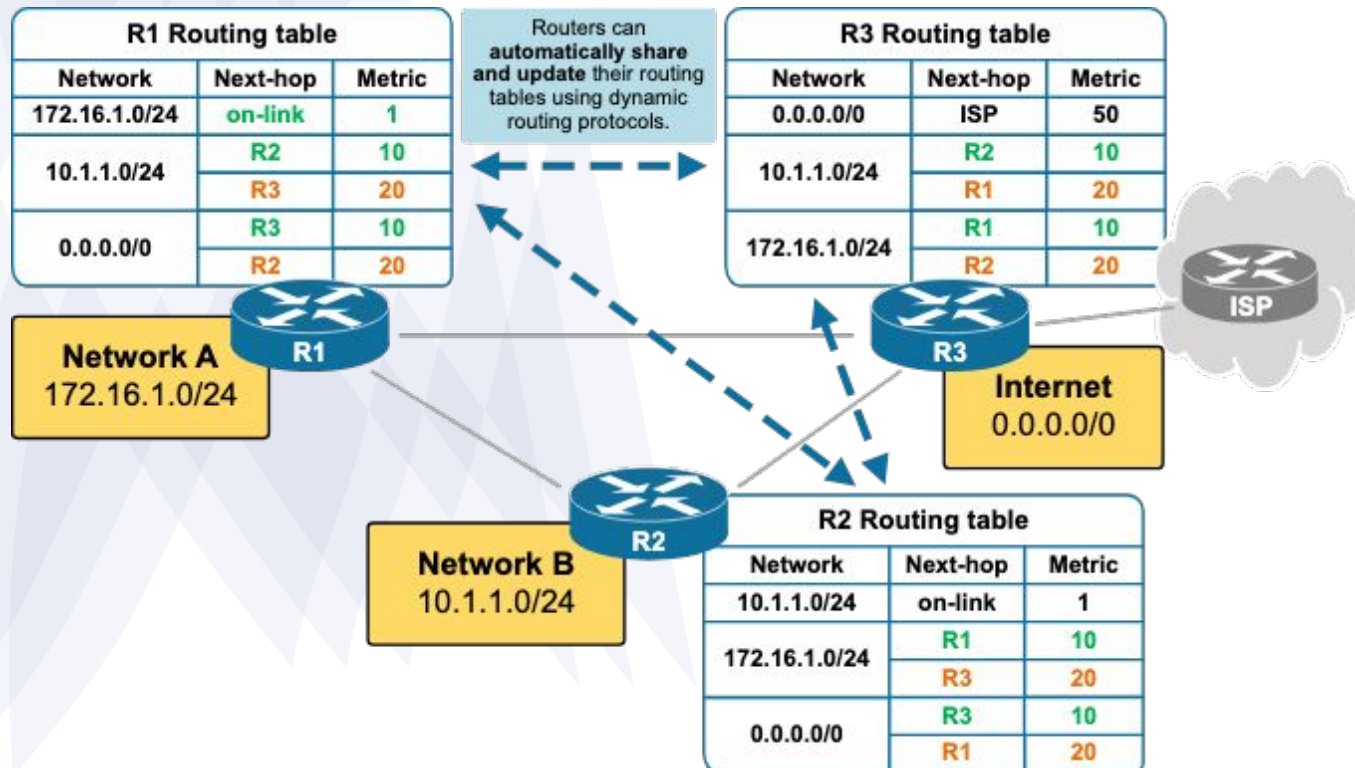


- Consider the scenario, three routers, each connected to a different network with their routing tables. Notice that each router can reach each network via two different paths.
- For example, R1 can reach 10.1.1.0/24 via R2 and via R3. However, R1 directly connects to 10.1.1.0/24 via R2 because it is the best path (has lowest metric).

IP Routing



- A router can build its routing table using two different methods:
 - Static routing is when an administrator manually configures routes. The router uses these fixed routes to forward packets, and they do not change unless the administrator updates them. This method is predictable and straightforward, but requires manual changes if the network topology changes.
 - Dynamic routing uses routing protocols to learn and update routes automatically. Routers exchange information about network paths and choose the best routes based on metrics such as distance, cost, or speed. This makes dynamic routing more adaptable and efficient in larger or constantly changing networks.



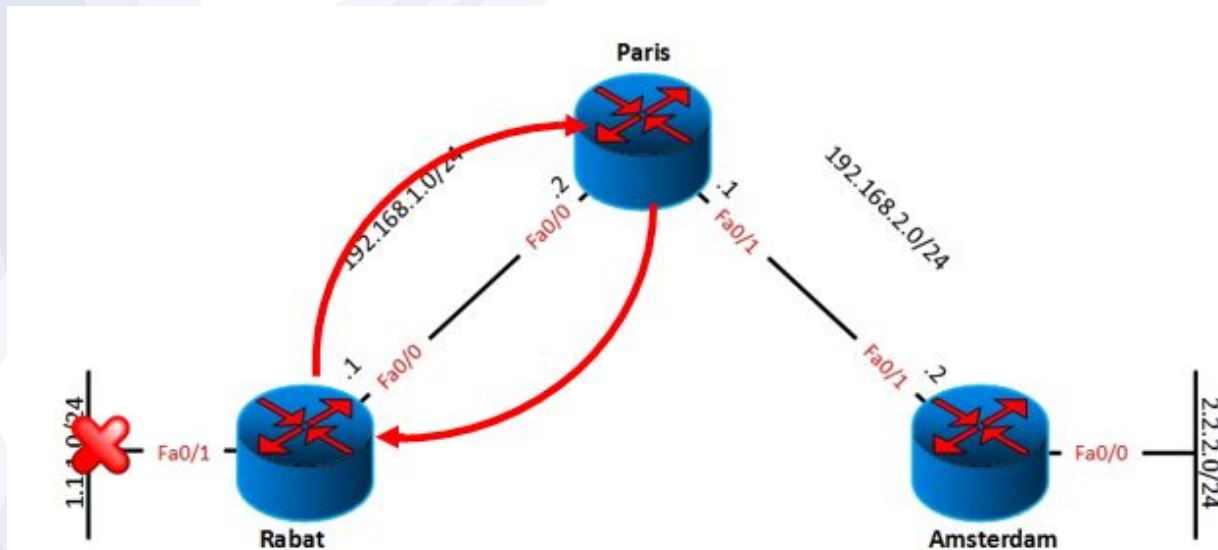
What is a routing algorithm?

- A **Routing Algorithm** is a method or procedure, or formula used by a routing protocol to calculate or determine the best path or route for data transfer
- A **routing algorithm** is a set of step-by-step operations used to direct network traffic efficiently
- When a packet leaves its source, there are could be many different paths it can take to its destination
- The routing algorithm is used to determine mathematically the best path to take
- The algorithm should lead to **consistent routing** i.e. the packets should be forwarded without **loops**

What is a routing algorithm?

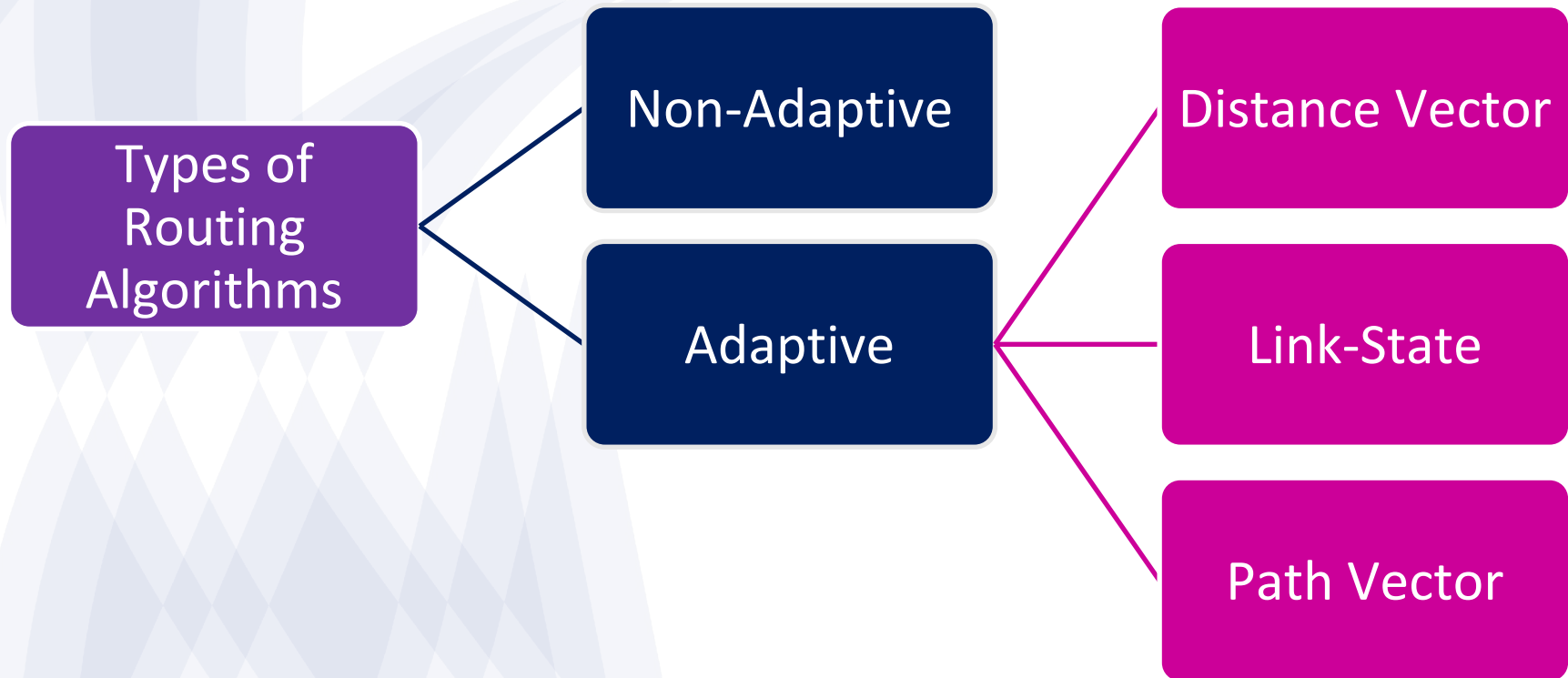


- A **loop** occurs when a packet is routed from one node to another node that could send back the packet to the previous node



- This back-and-forth forwarding could occur endlessly with the packet not being delivered
- Routing loops consume a lot of resources: CPU time and bandwidth

Categories of Routing Algorithms



Adaptive & Non-Adaptive Routing Algorithms



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- Non-Adaptive Routing Algorithms
 - Also known as a **static routing algorithms**
 - When booting up the network, the routing information is stored on the routers
 - They do not make routing decisions based on the network topology or network traffic
- Adaptive Routing Algorithms
 - Also known as **dynamic routing algorithms**
 - They make routing decisions based on the topology and network traffic
 - The main parameters related to this algorithm are hop count, distance and estimated transit time

Routing Metrics

- Determining a router's best path involves the evaluation of multiple paths to the same destination network and selecting the optimum or "shortest" path to reach that network
- The best path is selected by a routing protocol based on the **value** or **metric** it uses to determine the distance to reach a network
 - The **lower the metric the preferable the path**
 - **Note:**
 - the **routing algorithm** determines how the best path is computed while
 - the **routing metric** is the parameter used in the computation

- A **routing metric** is a value used by a routing protocol to determine which routes are better than others.
 - i.e. it is a calculated value used to determine the best path to a destination
- Metrics used in IP networks:
 1. Bandwidth
 2. Cost
 3. Delay
 4. Hop count
 5. Load
 6. Reliability

1. **Hop count** is the number of routers between a router and the destination network. Each router is equal to one hop.
 - **Example:** a router will prefer a path that is 3 hops away over a path that is 6 hops away
2. **Bandwidth** is the data carrying capacity of a link, sometimes referred to as the speed of the link.
 - The route with an accumulation of links that have the highest bandwidth values, or the fastest links would be a preferable path
3. **Load** is the amount of traffic on a certain link
4. **Delay** is the time a packet takes to traverse a path

5. **Reliability** is the probability of a link failure, calculated from the interface error count or previous link failures
6. **Cost** is a value determined either by the router's OS or by the network administrator to indicate preference for a route.
 - Cost can represent a metric, a combination of metrics or a policy

Static and Dynamic Routing



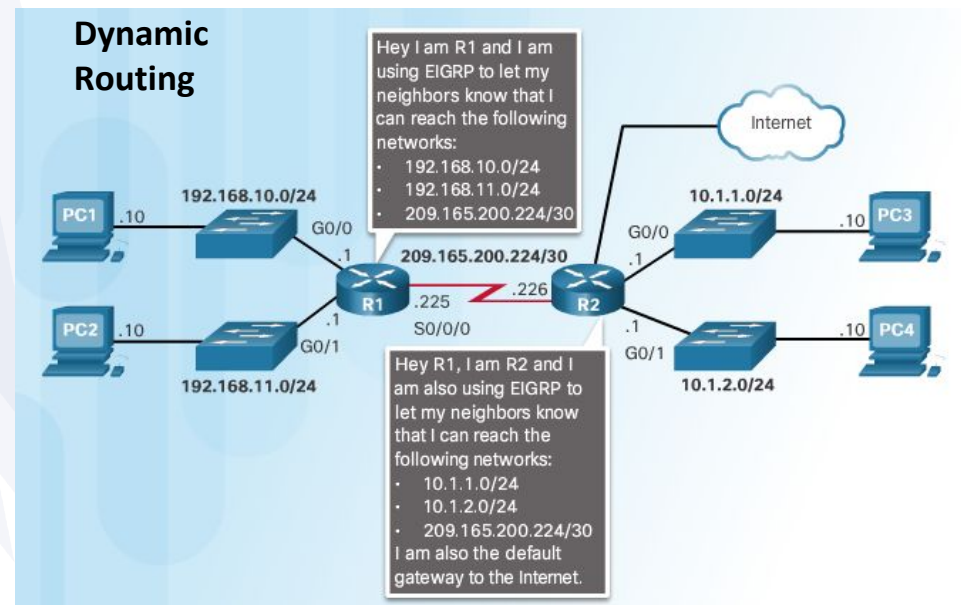
Routers learn about remote networks in two ways:

1. Statically – by manual configuration

- Static routes are not automatically updated and must be reconfigured when topology changes
- They use a non-adaptive routing algorithm

2. Dynamically

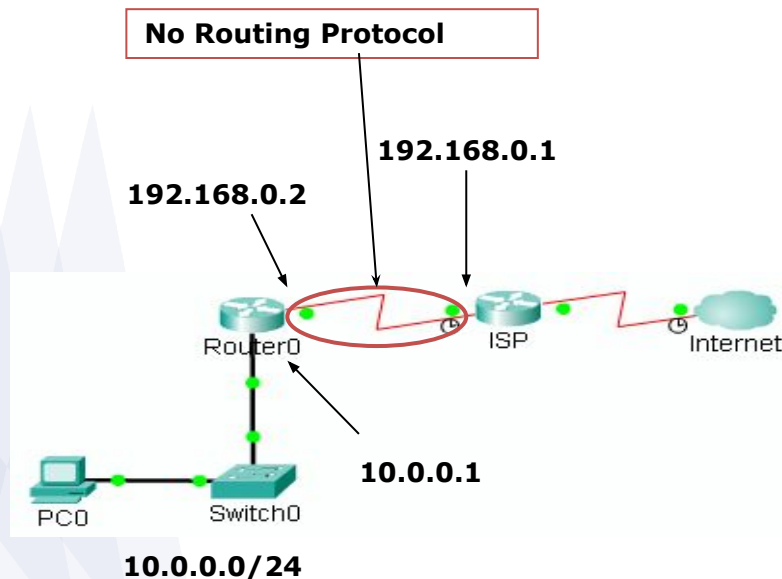
- They learn remote networks using adaptive algorithms



- A **static route** is simply a route that is added **manually** using a configuration command in a router
- After it is configured, IOS adds the route, including details such as the subnet address, mask, outgoing interface, and (or) next-hop router, into a new entry in that router's IP routing table
- After it is added, the router can then route packets whose destination IP address matches the static route
 - ◆ **NB:** The connected interfaces must be reachable for the remote network to be reached through the static route

Static routes are ideal for:

1. Small networks that are not expected to grow
2. Where a single default route is used to represent a path to any network not found in the routing table

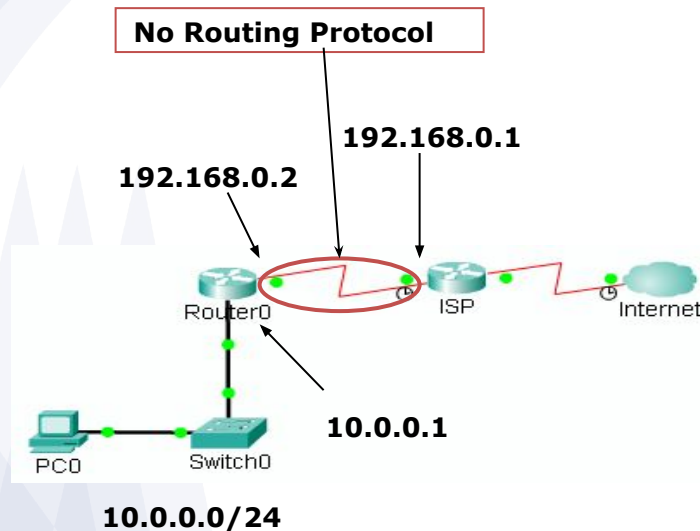


Example: Use default route on Router0 to point to ISP for all other networks beyond Router0

Static routes are ideal for:

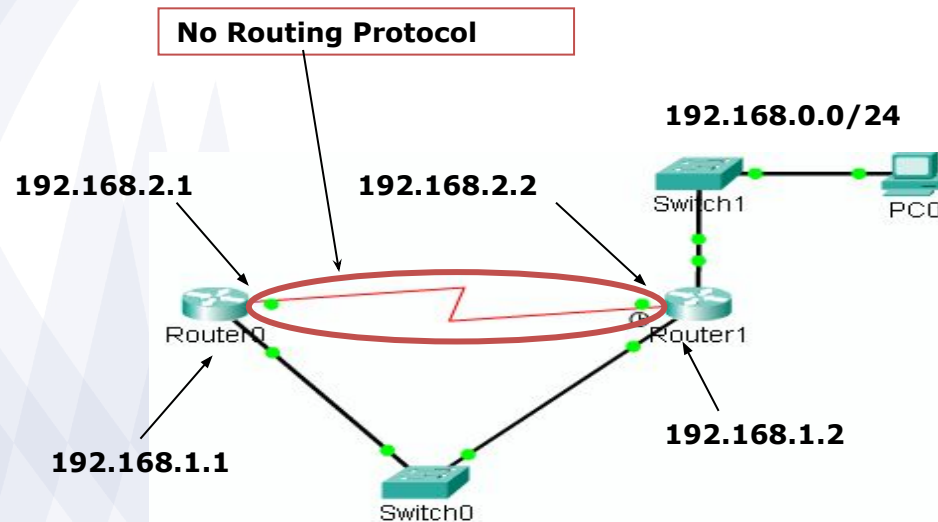
3. Routing to and from stub networks

- Stub network accessed by a single route and has one neighbor
- In the topology: 10.0.0.0/24 is a stub network



Example of a stub network where the link between ISP and Router0 is configured using a static method instead of using a routing protocol

R0 can reach the networks on R1 through the switch using a configured protocol. However, in case the route through the switch fails, then a static route can be configured between R0 and R1 as a backup route



4. Backup of other routes for use say when a dynamic route fails

1. Standard Static Routes

- consist of the destination network address and network mask, and the IP address of the next-hop gateway or exit interface
- you can configure multiple standard static routes with the same metric for load sharing

2. Default Static Routes

- a route that matches all packets
- The configuration has all zeros in the IP address and all zeros in the subnet mask. i.e. in IPv4 - 0.0.0.0/0 while in IPv6 - :::/0
- It identifies the gateway IP address to which the router sends all IP packets that it does not have in its routing table

3. Summary Static Routes

- A route that represents an aggregation of various network gateways
- The multiple static routes in the aggregation all use the same exit interface or next-hop IP address

4. Floating Static Routes

- routes that are used to provide a backup path to a primary static or dynamic route, in the event of a link failure
- the floating static route is configured with a higher administrative distance than the primary route so that it is not the preferred route but the one the router will resort to if when the preferred route is unavailable

Advantages of Static Routing



1. Can backup multiple interfaces/networks on a router for use say when a dynamic route fails
2. Easy to configure
3. No extra resources are needed (RAM, powerful CPU, bandwidth)
4. More secure - the path a packet takes between two destinations is always known precisely, and can be controlled exactly
5. Predictability- the network administrator computes the routing table in advance, the path a packet takes between two destinations is always known precisely, and can be controlled exactly
6. Ideal for small networks

Disadvantages of Static Routing



1. Initial configuration and maintenance is time-consuming
2. Configuration is error-prone, especially in large networks
3. Administrator intervention is required to maintain changing route information
4. Does not scale well with growing networks; maintenance becomes cumbersome
5. Requires complete knowledge of the whole network for proper implementation

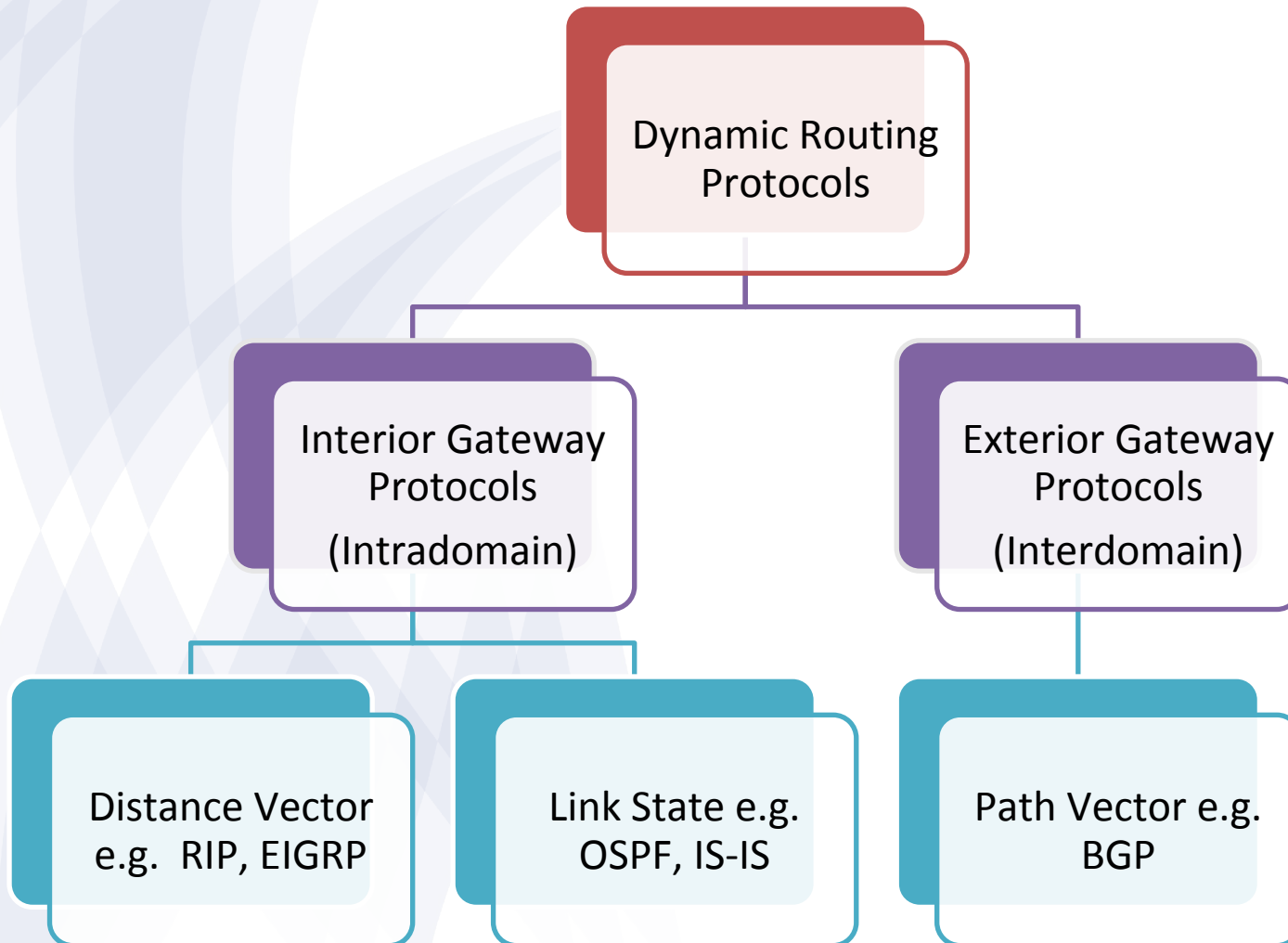
- Here a network routing protocol is configured e.g. RIP, OSPF, EIGRP, IS-IS, BGP etc
- Dynamic routing protocols use adaptive algorithms i.e. they adjust routing automatically when the topology or traffic changes
- Dynamic routing protocols will:
 - i. Discover remote networks and dynamically share this information with other routers
 - ii. Maintaining up-to-date routing information
 - iii. Choosing the best path to destination networks
 - iv. Find a new best path if the current path is no longer available

- Despite of their classification, all dynamic routing protocols have three main components:
 - i. **Data structures**
 - How data is organised and stored. Some routing protocols use tables and/or databases for their operations
 - is a way of collecting and organizing **data** such that we can perform operations on these **data** in an effective way
 - This information is stored in RAM
 - ii. **Routing protocol messages**
 - Communication exchanged between routers to allow them to gather information about the routing domain
 - i.e. they are used to discover neighboring routers, exchange routing information and maintain accurate information about the network
 - iii. **Routing Algorithms**
 - It is a method or procedure or formula used by a routing protocol to calculate or determine the best path for data transfer
 - Examples: DUAL, SPF, Bellman Ford's Algorithm

Categories of Dynamic Routing Protocols



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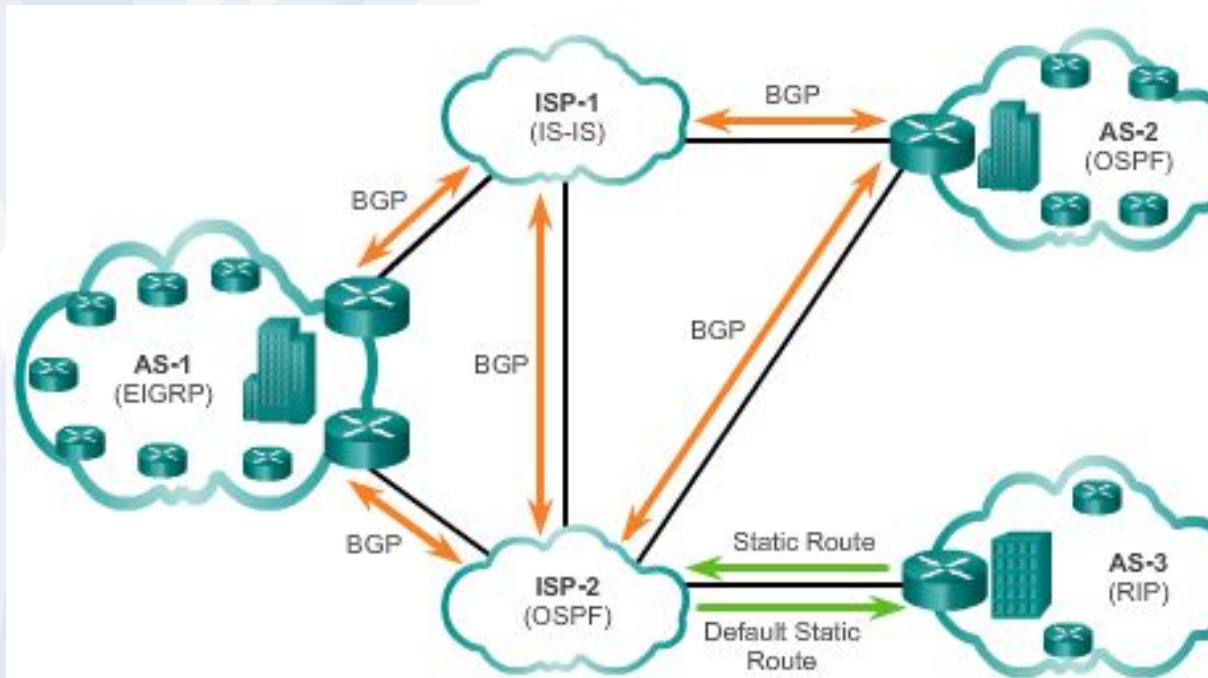


- Interior Gateway Protocols are used for routing inside an autonomous system & used to route within the individual networks themselves
 - Also known as **intradomain routing** protocols
 - Examples: RIP, EIGRP, OSPF, IS-IS
- Exterior Gateway Protocols (EGP)- Used for routing between autonomous systems
 - Also known as **interdomain routing** protocols
 - Example: BGPv4

IGP and EGP Routing Protocols



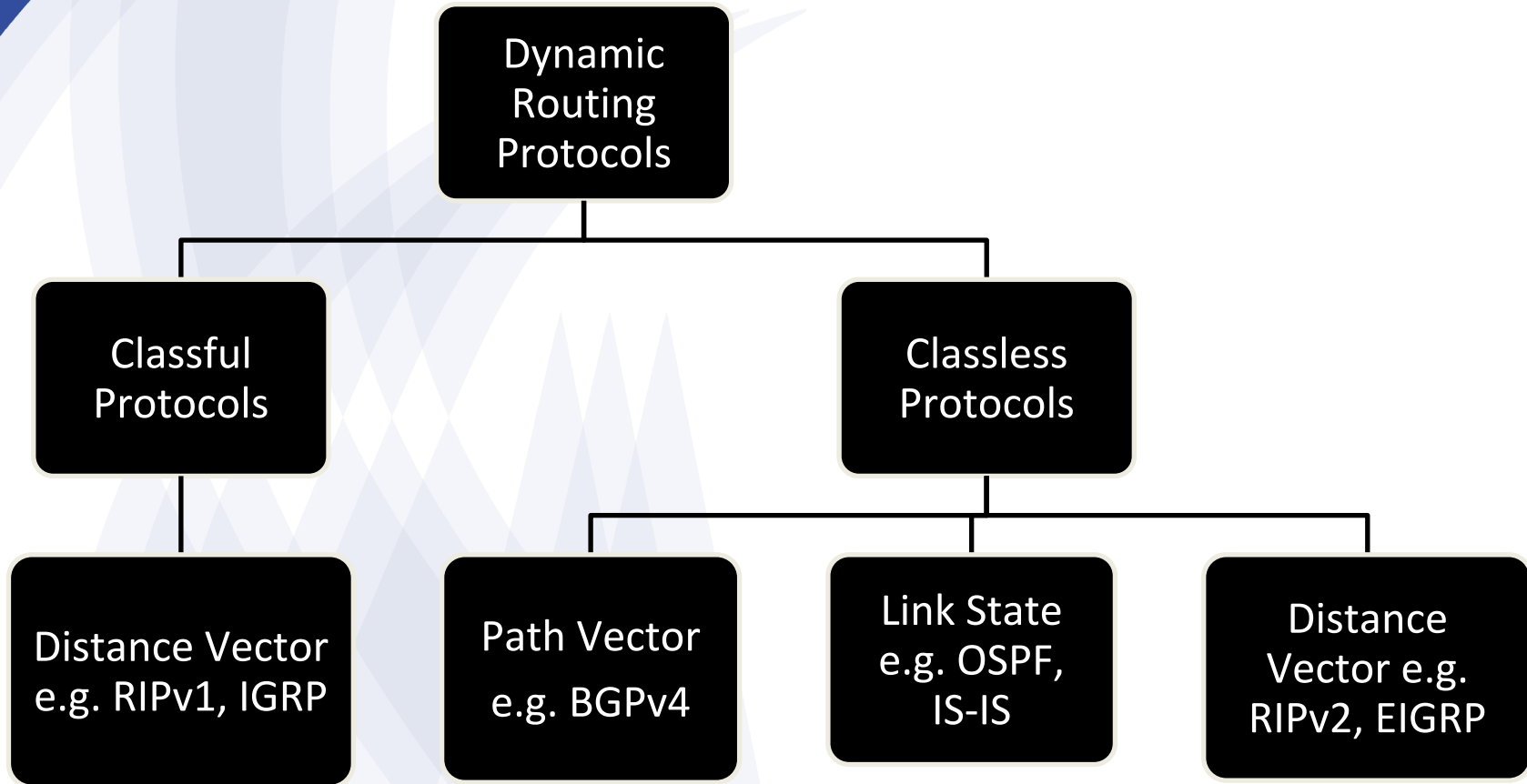
- An **autonomous System** is a group of routers under the control of a single authority
- Also sometimes referred to as a **routing domain**



Categories of Dynamic Routing Protocols



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Classful and Classless Routing Protocols



- Classful routing protocols do not include a subnet mask in the routing update
 - The receiving interface router interface has to determine what subnet mask to apply to the route (network) when placing in the routing table
 - Not good to configure in topologies that have VLSM, CIDR or are said to be discontinuous
- Classless routing protocols include a subnet mask in the routing update
 - They support VLSM, CIDR and can be configured where the network is said to be discontinuous

Advantages of dynamic routing

1. Prevents the time-consuming and laborious process of configuring static routes
2. Dynamically adapt to network changes making them give more current information
3. Scale better compared to static routing protocols
4. Less prone to configuration errors in scaling networks compared to static routing

Disadvantages of dynamic routing

5. Require more resources i.e. CPU time, interface bandwidth and RAM
6. Some protocols are complex and require expertise to configure and troubleshoot
7. Less secure. Additional configuration settings are needed to secure
8. Routes depend on current topology i.e. they can change